

Artifact Description

The artifact I selected for this enhancement is the same I have been using throughout this course, the Security Threat Monitoring Dashboard, a full-stack application developed using Node.js, Express, MongoDB Atlas, and JavaScript. The application allows authenticated users to create security logs, automatically generate alerts, and monitor security events through a dashboard interface. The focus of this milestone was improving the database layer through schema optimization, indexing, aggregation queries, and dashboard analytics.

Database Enhancement

The database enhancement expanded the original MongoDB implementation by using indexed collections, optimized schema design, and aggregation pipelines. New analytics summarize log data by severity level and event type while also ranking suspicious IP addresses based on cumulative threat scores. These enhancements improve both performance and usability while demonstrating practical database design concepts commonly used in security monitoring systems.

Course Outcomes

This enhancement shows the ability to design efficient database solutions, utilize modern database technologies, optimize query performance, and integrate backend data processing with a frontend dashboard. The project now much better reflects professional full stack development practices and showcases practical experience with MongoDB Atlas, Mongoose, aggregation pipelines, indexing, and RESTful API development.

Reflection

Completing this enhancement strengthened my understanding of how database design directly impacts application performance and scalability. Implementing indexes and aggregation queries showed how large datasets can be efficiently summarized into meaningful analytics for users. The project evolved from a basic CRUD application into a more realistic security monitoring dashboard capable of supporting future cloud deployment and advanced security analytics.